

Model Comparison: Additive vs Interaction

Additive Model

$$\theta_{jk} = \sum_c \beta_c \cdot (I_{kc} - I_{jc})$$

Effects combine linearly

Example: $A+B = \beta_A + \beta_B$

Maximum power Simple interpretation

Interaction Model

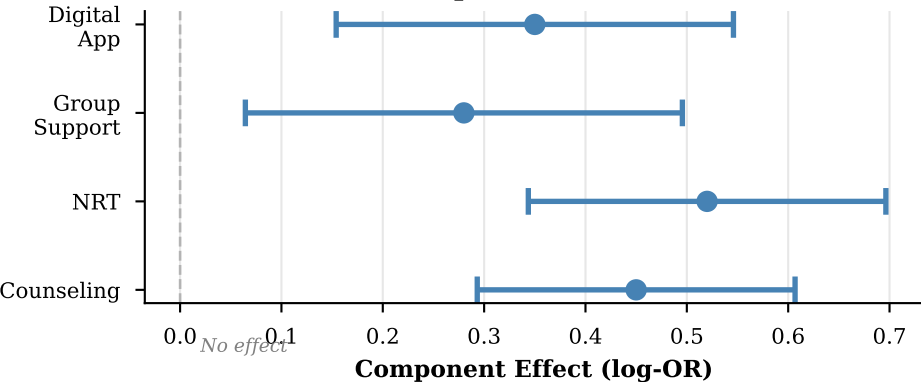
$$\theta_{jk} = \sum_c \beta_c \cdot \Delta I_c + \sum_{c < c'} \gamma_{cc'} \cdot \Delta I_c I_{c'}$$

Accounts for synergy/antagonism

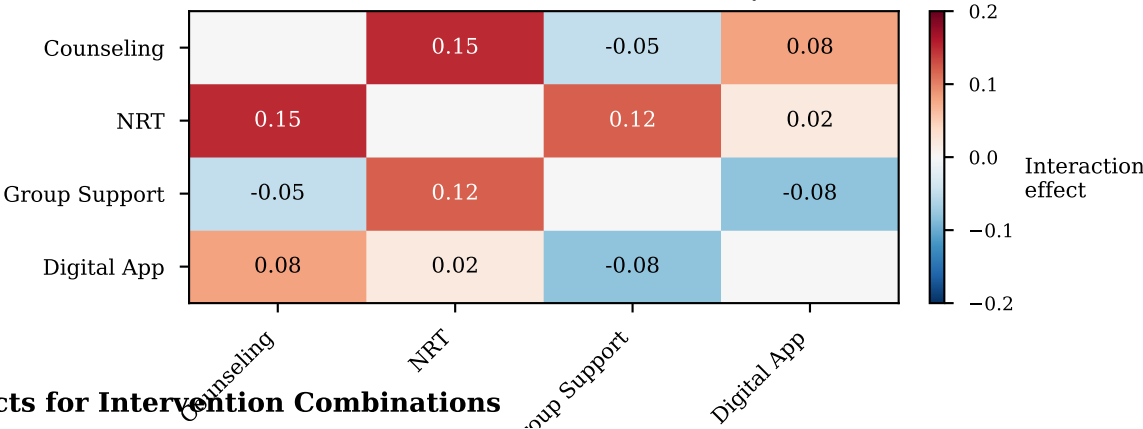
Example: $A+B = \beta_A + \beta_B + \gamma_{AB}$

Flexible Requires more data

Panel B: Component Effect Estimates



Panel C: Interaction Effects (γ)



Panel D: Predicted Effects for Intervention Combinations

